

Changho Choi

Seoul, Republic of Korea

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🎓 Google Scholar

Personal Profile

Computer Science undergraduate at Korea University specializing in computer vision and generative models, with 6+ years of applied machine learning experience in vision and audio. Actively seeking PhD opportunities in computer vision and general machine learning.

Education

Korea University

Seoul, Republic of Korea

Undergraduate

Mar 2017 - Present

- Major in Computer Science and Engineering
- Leave of absence for mandatory alternative military service (Nov 2021 – Oct 2024).

Gyeonggi Science High School

Gyeonggi-do, Republic of Korea

Graduated

Feb 2014 - Feb 2017

- Graduation thesis, "Eye Blinking Signal-Based Biometrics for Individual Identification Using Convolutional Neural Networks"

Research Experience

Vision & AI Lab

Seoul, Republic of Korea

Undergraduate Researcher

Mar 2025 - Dec 2025

- Conducted research on causal sequential visual encoders under Prof. Jinkyu Kim.
- Proposed MambaEye, a size-agnostic visual encoder that treats image recognition as a causal, sequential process, effectively mimicking human saccadic vision.

Publications

MambaEye: A Size-Agnostic Visual Encoder with Causal Sequential Processing

Changho Choi, Minho Kim, Jinkyu Kim

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings, 2026

SAO-Instruct: Free-form Audio Editing using Natural Language Instructions

Michael Ungersböck, Florian Grötschla, Luca A Lanzendörfer, June Young Yi, Changho Choi, Roger Wattenhofer

Poster presented at NeurIPS, 2025

LatentSwap: An Efficient Latent Code Mapping Framework for Face Swapping

Changho Choi*, Minho Kim*, Junhyeok Lee, Hyoung-Kyu Song, Younggeun Kim, Seungryong Kim

Arxiv Preprint, 2024

Work Experience

OptimizerAI

Seoul, Republic of Korea

AI Researcher

Oct 2024 - Dec 2024

- Contributed to research and development for the Text-to-SFX Model v2.
- Achieved a 62% human preference win rate, significantly surpassing ElevenLabs (25%) and the leading open-source model (13%).
- Attained a superior overall quality and diversity score of 7.67, beating the next-best competitor's score of 5.29.
- Performed literature reviews, data preprocessing, neural audio codec, and audio generation model training as part of the version 2 development process.

Pion Corporation

Seoul, Republic of Korea

AI Engineer

Jun 2023 - Oct 2024

- Developed a deep learning model, trained on in-service user behavior data, to recommend optimal product images, which replaced a legacy rule-based system and improved selection efficiency.
- Enhanced online product image clustering by integrating CLIP with advanced deep learning algorithms, utilizing Triton Inference Server and TensorRT for optimized performance.
- Created an API server using Beeble AI and ThreeJS for transforming images into relighted videos via FastAPI.

Maum.AI

Gyeonggi-do, Republic of Korea

AI Scientist

Mar 2019 - May 2023

- Experienced various computer vision tasks such as Super Resolution, JPEG noise restoration, Video Frame Interpolation, Face Swapping, Face Generation and Virtual Try-On through extensive literature review, utilization of open-source resources, and paper implementation.
- Contributed to two open-source projects resulted from research conducted for the company's products.
- Co-authored a research paper in collaboration with colleagues from the company's industrial project
- Led the computer vision research team, shaping research directions and managing the recruitment process.

Open-Source Projects

LatentSwap

Stars: 25

<https://github.com/usingcolor/LatentSwap>

Mar 2025

- Official PyTorch implementation for LatentSwap

HifiFace

Stars: 300+

<https://github.com/maum-ai/hififace>

Dec 2021

- Developed an unofficial PyTorch implementation of HifiFace, the state-of-the-art face swapping model at the time.

FaceShifter

Stars: 600+

<https://github.com/maum-ai/faceshifter>

Oct 2020

- Implemented FaceShifter, one of the first GAN-based face swapping models, from scratch in PyTorch.

Honors and Awards

ElevenLabs Worldwide Hackathon, Seoul

1st Prize

Feb 2025

- Voice Guardian: AI voice-based home protection system. Team members: Yooji Kwon, Jaemin Kim, Junyoung Lee, and Changho Choi.

Science Talent Scholarship

Full tuition scholarship awarded to selected students, Korea University

Feb 2017

- Part of SW-Centered University Project.